

Higher Diploma in Software Engineering (HDSE)

OFFICIAL ePROJECT DOCUMENTATION REPORT

VMS — ADVANCED IMMUNIZATION SYSTEM

Vaccination Management System

Institution: Aptech Computer Education (S.I.T.E. Center)

Course: Higher Diploma in Software Engineering

Academic Year: 2026

Document: Final eProject Documentation (v1.0)

PHP 8 MVC

MySQL

Tailwind CSS

Live Deployed

Submitted by Group: Batch 2508E — VMS Group

1. Team Members

This eProject has been jointly designed, developed, tested, and deployed by the following six enrolled students of the Higher Diploma in Software Engineering programme:

#	Student Name	Student ID	Contribution
1.	Abdul Ghaffar	1693661	Team Lead / Core MVC
2.	Muhammad Hamza	1697448	Backend & Database
3.	Haider Sultan	1693668	Frontend / UI
4.	Muhammad Imad	1697531	
5.	Shahaf Saeed Khan	1697533	
6.	Muhammad Sufyan		

Certificate of Completion

This is to certify that the eProject entitled “**VMS — Advanced Immunization System**” submitted by the above-named students is their own original work, carried out in partial fulfillment of the requirements for the **Higher Diploma in Software Engineering** at **Aptech Computer Education (S.I.T.E. Center)**, academic year 2026, under faculty supervision.

[Faculty Signature]

Project Supervisor

[Date]

Submission Date

2. Problem Definition

2.1 Background

Vaccination is the single most effective public health measure for preventing diseases such as polio, measles, hepatitis, and tuberculosis. In Pakistan, a significant share of immunization schedules — particularly those coordinated through schools and basic health units — is still managed with paper registers, manually updated charts, and verbal reminders. As coverage targets grow, this manual approach has become the weakest link in the immunization chain.

2.2 Problems with the Manual System

- **Missed Vaccination Opportunities:** Schedules are maintained on paper registers and notice boards. Parents and school staff frequently miss vaccination dates, leaving children with incomplete immunization cycles.
- **Time Delays and Man-Hours Lost:** Registering a child, verifying previous doses, scheduling the next dose, and updating records are all manual tasks that consume substantial man-hours of school administration and healthcare personnel.
- **Health Risks — Polio and Preventable Diseases:** Missed or delayed doses directly contribute to the spread of preventable diseases. In regions actively fighting polio, even one missed dose can compromise both the child's immunity and the community's eradication campaign.
- **Lost or Damaged Records:** Paper charts are prone to loss, damage, and duplication, making a child's complete immunization history difficult to retrieve when needed.
- **No Centralized Reporting:** With no unified database linking schools, parents, and hospitals, date-wise or area-wise coverage reports cannot be produced quickly, delaying administrative decisions.
- **No Notifications or Reminders:** Parents are not informed in advance of upcoming vaccination dates, so families without active school communication often fail to appear on schedule.

2.3 Proposed Solution — The e-Vaccination System

The **VMS — Advanced Immunization System** replaces the manual school vaccination process with a centralized web platform that:

- **Saves Man-Hours:** child registration, appointment scheduling, approvals, and status updates are digitized, eliminating paperwork and duplicate data entry.

- **Collects Infant Data:** every child's profile, dose history, and vaccination status is stored permanently in one database and can be retrieved in seconds.
- **Offers Alternative Opportunities for Vaccination:** when a scheduled date is missed, the parent can immediately search hospitals and book an alternative vaccination date, ensuring no child is left unvaccinated.
- **Produces Date-Wise Reports:** administrators can view upcoming vaccination dates and generate date-wise coverage reports in real time.
- **Notifies Parents:** dashboard notifications and reminders inform parents ahead of every due vaccination date.

3. System Modules (Customer Requirement Specification)

The system serves three actors. Their functional requirements are summarized below.

3.1 Administrator Module

Functionality	Description
View All Child Profiles	Browse and search complete profiles of every registered child in the system.
Upcoming Vaccination Dates	View a consolidated list/calendar of all upcoming scheduled vaccinations.
Date-Wise Reports	Generate and view vaccination reports filtered by date for analysis and coverage monitoring.
Check Vaccine Availability	Monitor real-time vaccine stock/availability across all affiliated hospitals.
Approve / Reject Requests	Review vaccination appointment requests booked by parents and approve or reject them.
Hospital Management	Add, update, delete, and list hospital details (name, address, location, contact).
Parent Booking Details	View all appointment bookings made by parents (child, hospital, vaccine, date, status).

3.2 Parent Module

Functionality	Description
Register & Login	Create an account and log in securely with session-based authentication.
Child Details Maintenance	Add, update, and maintain complete details of every child (name, DOB, gender, blood group).
Dashboard Notifications	Receive notifications on the dashboard for upcoming vaccination dates.
Search Hospitals & Book Schedules	Search affiliated hospitals and book a vaccination schedule by selecting child, hospital, vaccine, and date.

Reports of Taken Vaccinations	View reports/history of all vaccinations already taken (completed doses).
Manage Profile	Update personal profile information and password.

3.3 Hospital Module

Functionality	Description
Register & Login	Hospital is registered with name, address, and location; staff log in securely with issued credentials.
Receive Appointments	Receive vaccination appointments booked by parents through the admin/booking flow, with full child and vaccine details.
Update Vaccination Status	After completing the dose, update the child's status to “Vaccinated” or “Not Vaccinated” ; the system maintains a complete status history.

4. Hardware & Software Requirements

4.1 Hardware Requirements (Minimum)

Component	Requirement
Processor	Pentium 166 MHz or higher
RAM	128 MB or higher
Hard Disk	100 MB free space (application + database)
Network	Internet connectivity for hosted / live access

4.2 Software Requirements

Software	Requirement
Operating System	Linux or Windows 2000 Server (or any modern Linux/Windows distribution)
Web Server	Apache
Server-Side Language	PHP
Database	MySQL
Additional Support	PERL
Browser	Any standard web browser (Chrome / Firefox / Edge)

4.3 Actually Deployed Environment (for Evaluation)

Item	Value
Live Server	InfinityFree Shared Hosting — PHP 8.x, Apache-compatible
Database	MySQL 8 (hosted)
Frontend	Tailwind CSS (CDN) + Vanilla JavaScript
Architecture	Custom PHP MVC (front-controller, no framework), Argon2id/Bcrypt passwords, CSRF-protected forms, role-based access control

5. Login Credentials for Evaluation

Live URL	[LIVE URL – https://vams.infinityfreeapp.com/]
Application Path	/login (Sign In page)
Password Recovery	“Forgot password?” link on login page — single-use reset link, valid 1 hour

5.1 Administrator Credentials

Email	Password
admin@aku.edu.pk	password
usman.tariq@shifa.edu.pk	password

5.2 Parent Credentials

Email	Password
ali.raza@gmail.com	password
fatima.ahmed@outlook.com	password
hassan.malik@yahoo.com	password
saira.bibi@hotmail.com	password
omar.farooq@gmail.com	password

5.3 Hospital Credentials

Email	Password
info@aku.edu.pk (Aga Khan)	hospital123
info@shifa.edu.pk (Shifa International)	hospital123
info@jinnah.edu.pk (Jinnah Lahore)	hospital123
navyshifa@pns.net (PNS Shifa)	hospital123
nadia.iqbal@aku.edu.pk	password
imran.shah@shifa.edu.pk	password

sana.queshi@jinnah.edu.pk	password
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Note for Examiner: Open the Live URL → click **Sign In** → use any credential above. Each role redirects to its own dashboard (Admin → /admin/dashboard, Parent → /parent/dashboard, Hospital → /hospital/dashboard).

6. System Screenshots

The screenshots below were captured directly from the running **VMS — Advanced Immunization System** application during final evaluation testing.

6.1 Login Page & Password Recovery

Figure 6.1 — Login page (Sign In) with “Forgot password?” link

Figure 6.2 — Forgot password page — single-use reset link, valid for 1 hour

6.2 Administrator & Hospital Dashboards

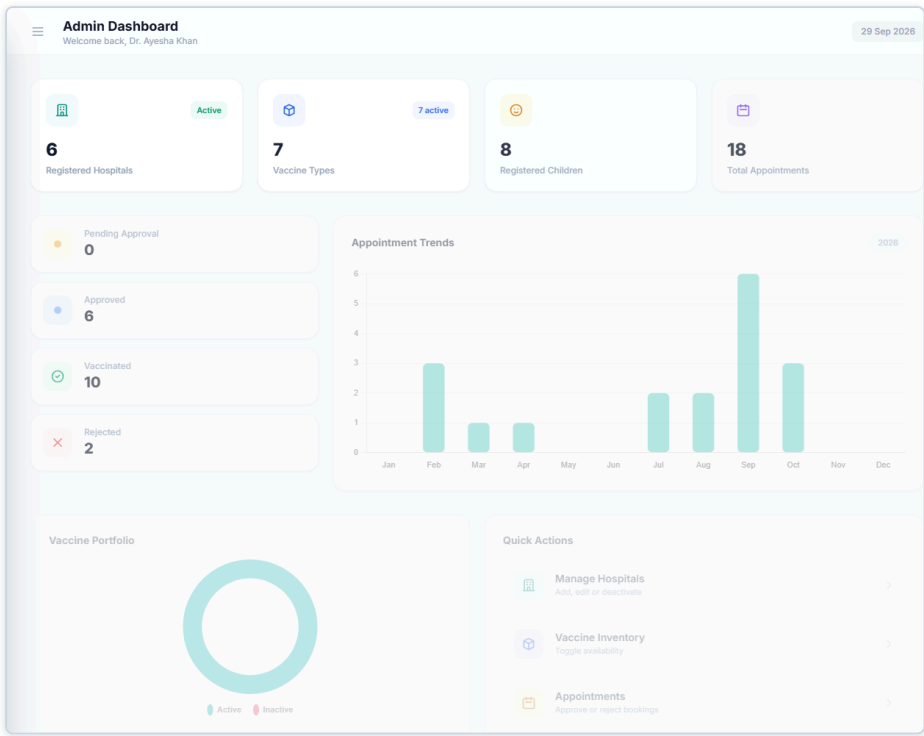


Figure 6.3 — Administrator dashboard — statistics, appointment trends, and quick actions

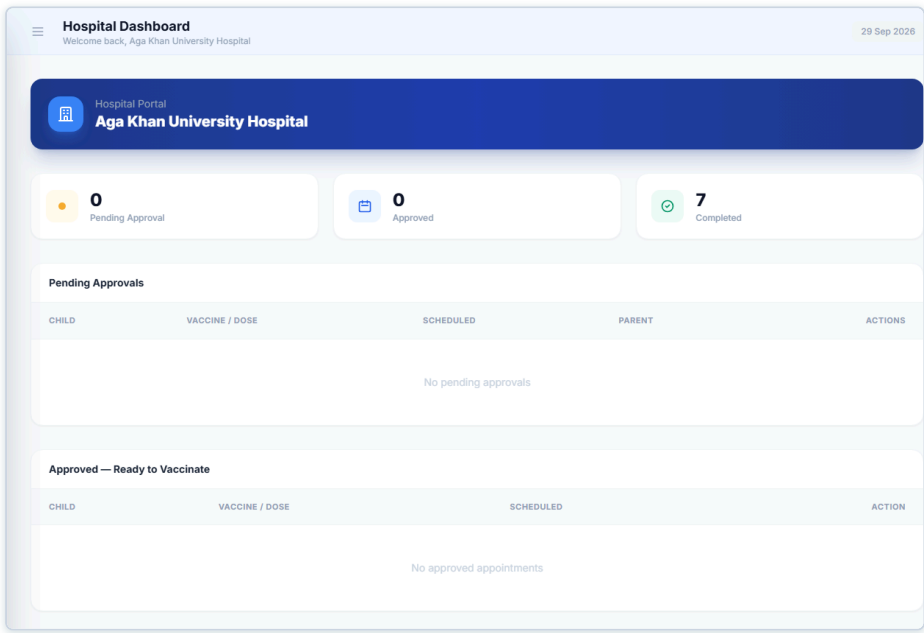


Figure 6.4 — Hospital dashboard — pending approvals and ready-to-vaccinate queue

6.3 Parent Dashboard & Vaccine Inventory

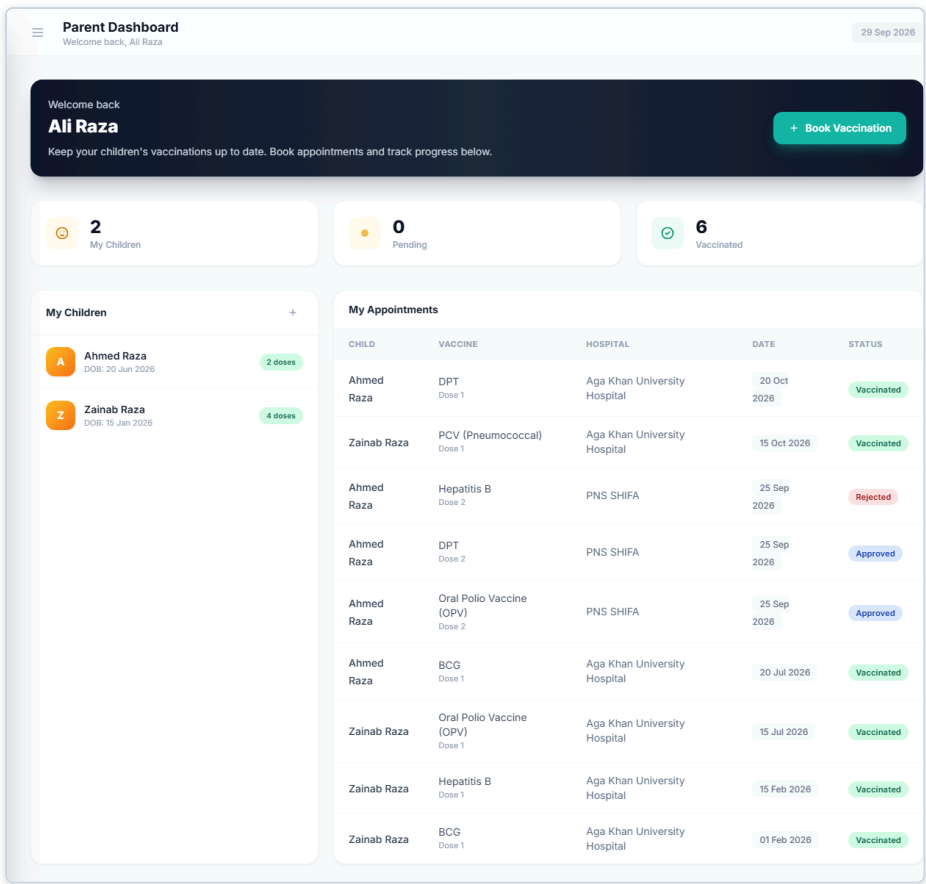


Figure 6.5 — Parent dashboard — children, appointments, and vaccination status

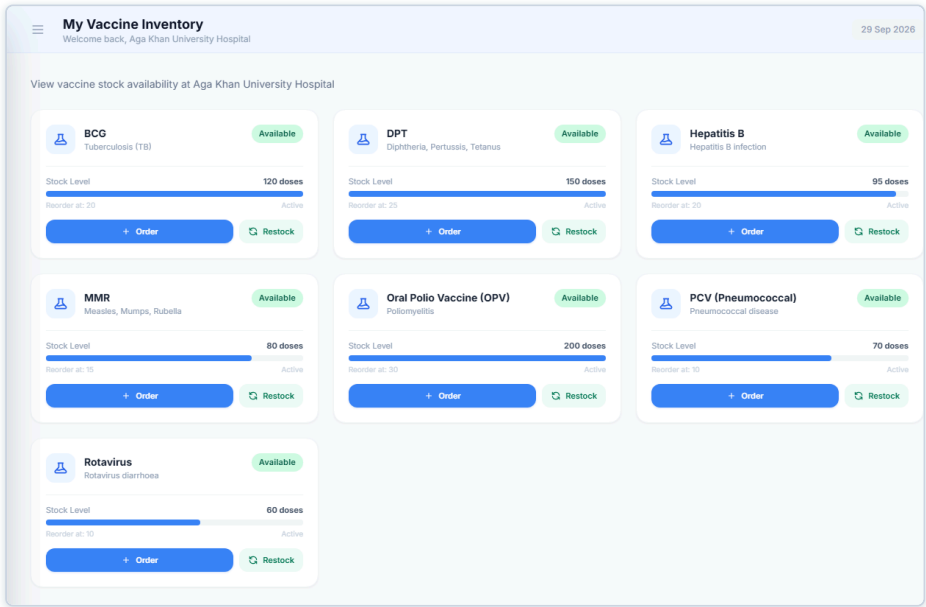


Figure 6.6 — Hospital vaccine inventory — stock levels with order & restock controls